of the five string formulations we were working with. We argued that the interchange of winding and vibration modes of a string allows us to rephrase exactly the string theoretic description of a universe with a circular dimension of radius 1/R in terms of one in which the radius is R. The point we glossed over is that the Type IIA and Type IIB string theories actually get exchanged by this duality, as do the Heterotic-O and Heterotic-E strings. That is, the more precise statement of the large/small radius duality is this: The physics of the Type IIA string in a universe with a circular dimension of radius R is absolutely identical to the physics of the Type IIB string in a universe with a circular dimension of radius 1/R (a similar statement holds for the Heterotic-E and Heterotic-O strings). This refinement of the large/small radius duality has no significant effect on the conclusions of Chapter 10, but it does have an important impact on the present discussion.

The reason is that by providing a link between the Type IIA and Type IIB string theories, as well as between the Heterotic-O and Heterotic-E theories, the large/small radius duality completes the web of connections, as illustrated by the dotted lines in Figure 12.10. This figure shows that all five string theories, together with M-theory, are dual to one another. They are all sewn together into a single theoretical framework; they provide five different approaches to describing one and the same underlying physics. For some or other application, one phrasing may be far more effective than another. For instance, it’s far easier to work with the weakly coupled Heterotic-O theory than it is to work with the strongly coupled Type I string. Nevertheless, they describe exactly the same physics.

The Overall Picture

We can now more fully understand the two figures—Figures 12.1 and 12.2—that we introduced in the beginning of this chapter to summarize the essential points. In Figure 12.1 we see that prior to 1995, without taking any dualities into account, we had five apparently distinct string theories. Various physicists worked on each, but without an understanding of the dualities they appeared to be different theories. Each of the theories had variable features such as the size of their coupling constant and the geometrical form and sizes of curled-up dimensions. The hope was (and still is) that these defining properties would be determined by the theory itself, but without the ability to